CS3751 Data Visualization

Class Project Report

Quiz Attempt Analysis: Hypothesis Testing Through Visualization

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1 Introduction

In modern digital learning, online quizzes are more than just a way to grade students; they provide a window into how students learn. By looking at how long students spend on a quiz, which questions they find difficult, and how they improve when they try again, we can find patterns that help improve education.

This report presents a data visualization study using over 31,000 quiz attempt records from a Moodle system. As part of the CS3751 Data Visualization module at the University of Moratuwa, this project aims to turn raw data into clear insights using effective design principles.

The analysis is divided into two main parts:

- **Task 1: Verification of Predefined Hypotheses.** We test five provided claims regarding time management, question difficulty, and the progress of high vs. low-performing students.
- **Task 2: Exploration of Novel Hypotheses.** We propose and verify five new questions to understand student retry behavior and score improvement across different quizzes.

Every hypothesis is tested using carefully chosen visualizations. We explain the choice of marks and channels for each plot to ensure they follow good design principles, leading to data-driven conclusions that either accept or reject each claim.

2 Background

The rise of Digital Learning Environments has transformed academic assessments into rich sources of behavioral data. Within Computer-Based Assessment (CBA), Learning Management Systems (LMS) like Moodle do not merely record success or failure; they capture the process of learning through granular event logs. This metadata including precise start times, total duration, and individual question performance allows for a deeper exploration of student engagement strategies that go beyond a simple final grade.

By analyzing this data, educational researchers can address several critical areas of assessment design:

- **Item-Level Diagnostics:** Evaluating individual question performance helps identify specific "cognitive hurdles." If certain questions consistently yield low marks, it suggests a gap in the curriculum or a need to refine the question's phrasing.
- **Temporal Dynamics:** Investigating the correlation between *time-on-task* and achievement helps identify the "Goldilocks Effect." This allows instructors to deter-

mine whether students are rushing through content (potentially guessing) or struggling with time constraints, leading to more equitable exam time limits.

- **Iterative Progress Tracking:** Since modern systems allow for multiple attempts, we can track "Mastery Learning." Analyzing whether students show a steady upward trajectory or a performance plateau provides evidence of whether they are learning from feedback or simply repeating attempts mechanically.

However, interpreting thousands of raw interaction records is a significant challenge for educators. This study utilizes Visual Analytics to bridge this gap. By mapping complex numeric and temporal variables to effective visual channels.Such as spatial position, length, and hue.we aim to uncover the underlying behavioral patterns of the student cohort. Ultimately, this report uses data visualization as a diagnostic tool to provide evidence based recommendations for improving instructional quality and assessment fairness.

3 Dataset

3.1 Source and Structure

The dataset was exported from a Moodle instance and contains attempt-level records for three quizzes (Quiz 1, Quiz 2, Quiz 3). Each quiz comprises five questions, each worth 2.00 marks, giving a maximum total grade of 10.00. The following attributes are included per attempt:

Table 1: Dataset Attributes

Attribute	Type	Description
Student Code	Integer	Anonymous student identifier
State	Categorical	`Finished` or `In progress`
Started on	Datetime	Attempt start timestamp
Completed	Datetime	Attempt end timestamp
Time taken	String	Duration in Moodle format (e.g. *7 mins 25 secs*)
Grade/10.00	Float	Total grade awarded
Q. 1–5 / 2.00	Float	Per-question mark (0, 1, or 2)
quiz	Integer	Quiz identifier (1, 2, 3)
time_minutes	Float	Derived: time taken in minutes
attempt_num	Integer	Attempt sequence number per student

3.2 Data Cleaning

The raw Moodle export required the following cleaning steps before analysis:

Step 1. Filter Finished Attempts. In progress records were excluded as they lack valid grade or completion timestamps required for temporal analysis.

Step 2. Numeric Coercion. Grades and individual question marks (e.g., Q1–Q5) were converted from string format to floating-point numbers.

Step 3. Temporal Normalization. Moodle time strings (e.g., *7 mins 25 secs*) were parsed using regular expressions to calculate total seconds, which were then converted to minutes for standardized scaling.

Step 4. Outlier Mitigation. Attempts exceeding a threshold of **7,200 seconds (2 hours)** were removed. Statistical verification showed that >**97%** of genuine attempts finish within this window; values beyond this (some reaching several days) represent cases where a browser session was left inactive.

Step 5. Attempt Sequencing. A chronological `attempt_num` column was generated for each student per quiz to allow for longitudinal progress tracking.

Step 6. Multi-Quiz Aggregation. Finally, the individual datasets for Quiz 1, 2, and 3 were merged into a single master dataset. A `quiz_id` column was added to preserve the source context, enabling comparative analysis across different assessment modules.

3.3 Final Dataset Summary

After cleaning, the dataset contains **31,334 attempt records** across the three quizzes, representing **up to 7 attempts** per student per quiz. Table 2 provides a per-quiz summary.

Table 2: Post-Cleaning Dataset Summary

Metric	Quiz 1	Quiz 2	Quiz 3
Mean grade	3.81	4.78	6.27
Median time (min)	5.47	6.20	5.87
Max attempts	7	6	5
Questions	5	5	5
Max marks	10.00	10.00	10.00

4 Methodology

Each hypothesis was evaluated using the following workflow:

1. **Data preparation.** Relevant subsets (quiz number, attempt filter, performance group) were extracted from the cleaned dataset.
2. **Visualization design.** Chart types were chosen based on the nature of the variables (continuous, ordinal, categorical) and the relationship being tested. Marks and channels were selected to maximise perceptual effectiveness following Munzner's visualization framework.
3. **Interpretation.** Each chart was examined for trends, patterns, or the absence thereof, and a verdict was drawn.

For Task 2, five additional hypotheses were formulated from exploratory patterns in retry counts, inter attempt waiting time, threshold attainment, and cross quiz first-attempt performance. In the retry gap analysis, impossible negative waiting times caused by timestamp inconsistencies were excluded before interpretation.

All visualizations were produced in Python using `matplotlib` and `seaborn`. The cleaned dataset (`cleaned_data.csv`) was loaded for all analysis.

5 Results

5.1 Hypothesis 1 : Students who take longer to complete the quiz tend to score higher

Conclusion: Rejected

The hypothesis is rejected based on the visual evidence. It is observed from the visualization that while there is a slight fluctuation in the averages of the grade, the grade distribution is almost same after 10 to 15 minutes. It is clearly shown that a student who spends 50 minutes on the quiz performs approximately the same on average as a student who spends 15 minutes. Therefore, it is concluded that taking extra time does not correspond to higher marks. Additionally, it is noted that in some boxes, the mean is pulled below the median line, indicating that a heavy tail of low scoring students is dragging the average down in those specific time brackets.

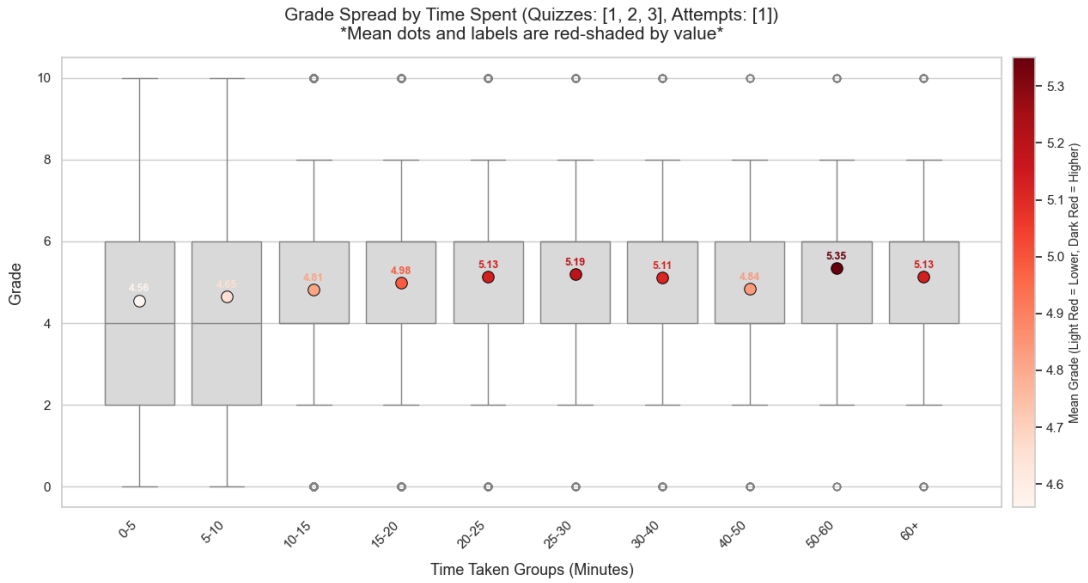

Figure 1: Grade spread by time spent, utilizing a grouped box plot with mean markers.

Choice of Visualization

A Box Plot with time bins is utilized because it effectively abstracts raw, dense data into clear statistical summaries such as medians, quartiles, and ranges. The statistical data is expressed through these structural marks appropriately. The visualization is made more effective by the addition of the mean markers. Instead of forcing the viewer to guess the trend, the visual system is allowed to simply connect the dots across the screen to instantly see how the trend goes.

To ensure the boxes are easily readable and clearly discriminated, the X-axis is structured carefully into 5-minute intervals for early attempts where most data lives and 10-minute intervals for later attempts, with X-axis labels rotated to prevent overlap. Furthermore, a subtle horizontal grid background is applied so that marks are visually separable. This ensures that boxes located far on the right side of the screen can be accurately traced back to the exact grade on the Y-axis without spatial distortion.

Visual Encoding Breakdown:

- **Marks:** Lines and areas are used to form the boxes, which represent the statistical distribution of the grades of the students with the median line, the middle 50% box, and outlier lines. Additionally, point marks are used inside the boxes to explicitly represent the average score.
- **Channels:**
  - *Spatial Position (Y-axis):* Used to encode the quantitative attribute of the grades.
  - *Spatial Position (X-axis):* Used to encode the time taken, which is categorized

into sequential bins.

- *Color:* A sequential color palette ranging from white to dark red, is applied specifically to the mean markers as a redundant encoding to map the ordered progression of averages from lowest to highest. A uniform gray color is used for all the boxes, ensuring they provide structural context without distracting from the color-coded trend of the mean markers.
- *Shape:* A circular shape is used to represent the point marks.

5.2 Hypothesis 2 : Some questions are consistently harder than others

Conclusion: Accepted

The hypothesis is accepted based on the visual evidence. It is observed from the visualization that certain questions consistently show a significantly lower average score compared to others in multiple attempts. For example, the bars for question 4 in quiz 1 and question 3 in quiz 3, remain noticeably lower than the other bars of that particular quiz across all attempts. Therefore, it is concluded that specific questions are consistently more difficult for the students to answer correctly.

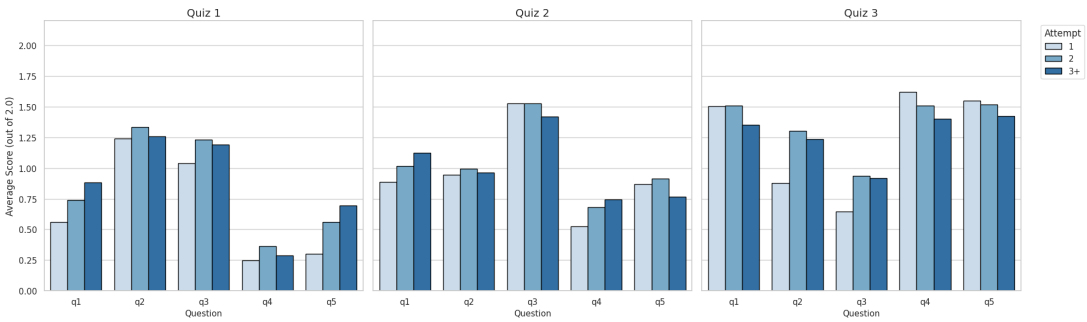

Figure 2: Average score per question across attempts, utilizing a single grouped bar chart.

Choice of Visualization

A Grouped Bar Chart is utilized because it effectively allows for the comparison of average scores for multiple questions across different attempts on a single, unified graph. The visualization is made more effective by utilizing spatial alignment on a common scale. Because all bars share the same baseline on the Y-axis, accurate relative judgments can be instantly made. To ensure the visual encoding accurately matches the data type, a sequential colormap consisting of varying shades of blue is utilized. This appropriately represents the ordered, ordinal sequence of the attempt numbers, rather than using unrelated random colors. Furthermore, the bars for attempts 1, 2, and 3+ for a single question are grouped right next to each other and because of that, they are instantly perceived as a related group, allowing for quick detection of score changes across attempts without much effort. Also, to maintain clear discriminability and separability, a black border is

added to each bar, ensuring that every mark remains distinctly separable from the white background and its adjacent bars.

Visual Encoding Breakdown:

- **Marks:** 1D Rectangles/Bars are used as the main visual mark.
- **Channels:**
  - *Spatial Position (Y-axis):* Used to encode the quantitative attribute of the average score.
  - *Spatial Position (X-axis):* Used to encode the categorical identity of the questions.
  - *Color :* A sequential color channel which shows varying shades of blue, is used to map the ordered sequence of the attempt numbers.

5.3 Hypothesis 3 : High-performing students consistently improve scores over multiple attempts, whereas low performers show erratic progress

Conclusion: Rejected

The hypothesis is rejected based on the visual evidence. To test this hypothesis, it was assumed that students scoring 8-10 points (4-5 correct questions) on their first attempt were "High Performers," and students scoring 0-4 points (0-2 correct questions) were "Low Performers." The visualization reveals that high performers actually crash massively on their second attempt, represented by a sharp downward trend. On the other hand, low performers demonstrate some erratic improvement across their subsequent attempts.

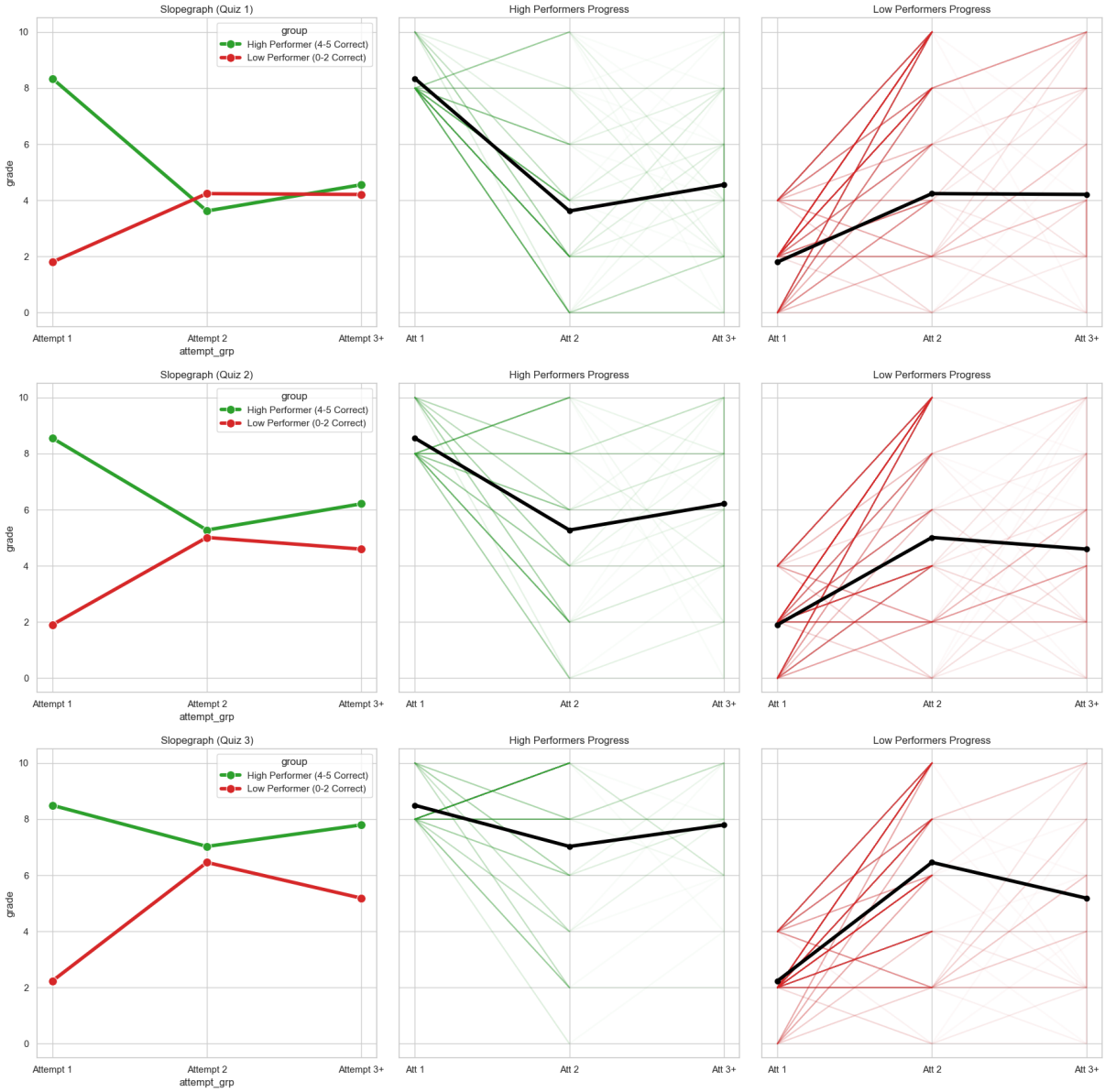

Figure 3: Progress trend by performance cohort, utilizing a faceted juxtaposed layout with slopegraphs and small multiples.

Choice of Visualization

A Faceted Juxtaposed Layout combining a Slopegraph and Small Multiples with Spaghetti Lines is used here, because it effectively manages visual complexity while allowing for comparison across student cohorts. To handle complexity without violating the Discriminability and Separability principles, the data is faceted and juxtaposed into three side-by-side panels. If hundreds of lines were plotted on a single graph, it would result in a visually incomprehensible smudge. Furthermore, individual student lines are rendered as a transparent background texture, and one clear black Mean Line is superimposed on top. This maintains a low cognitive load, allowing the exact mathematical average to be seen instantly while still providing the faded background lines to judge individual student

behavior.

Visual Encoding Breakdown:

- **Marks:** 1D Connection Lines are used as the foundational mark. The points joined by a line are instantly recognized as belonging to the same student, which successfully creates a visual perception of progress of that particular student.
- **Channels:**
  - *Spatial Position (Y-axis):* Used to encode the quantitative attribute of the grades.
  - *Spatial Position (X-axis):* Used to encode the sequential, ordinal data of the attempt number (Attempt 1, Attempt 2, Attempt 3+).
  - *Color:* Used to encode categorical identity. Green and red are used to represent High Performers and Low Performers respectively.
  - *Color Opacity:* Used to encode density. The individual student spaghetti lines are set to transparent for some extend. The color naturally darkens, where lines overlap which creates a derived visual encoding of density that clearly highlights large groupings of students following the same trend.

5.4 Hypothesis 4 : More difficult questions take longer to answer, but higher-performing students answer them faster than lower-performing students

Conclusion: Rejected

The hypothesis is rejected based on the visual and statistical evidence. Because the dataset only records the total time taken for the entire quiz rather than the time spent on individual questions, total quiz time was utilized as an alternative. To conduct this analysis, several assumptions were made which are : the hardest question was identified mathematically as the question with the lowest average score; students were categorized as High Performers (4–5 correct) or Low Performers (0–2 correct) based strictly on their first attempt; and the data was filtered to include only students who got the hardest question completely correct. This filtering ensures the time measured reflects actual problem solving effort rather than rapid, incorrect guessing. The visualization reveals these details. In quiz 1, high performing students completed the assessment faster than lower-performing students. However, in quiz 2 and 3, high performers actually took slightly longer time compared to low performing students. Since 2 out of 3 quizes show results that contradicts with the hypothesis, it was rejected.

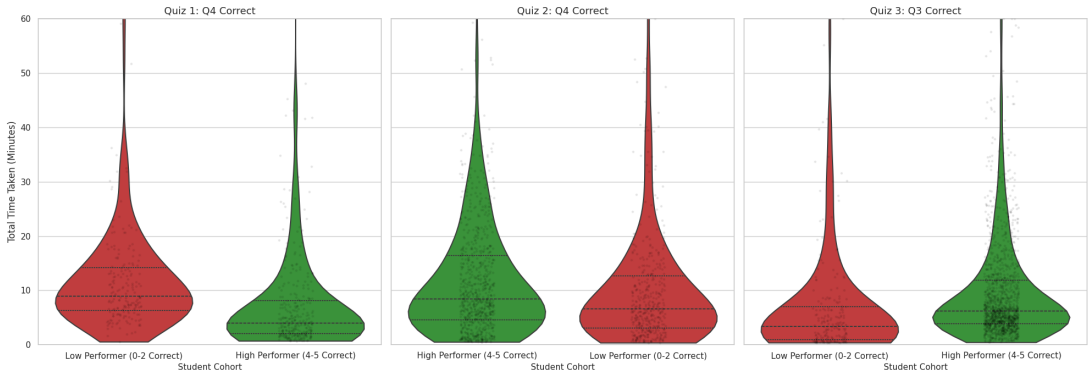

Figure 4: Total quiz time distribution for students answering the hardest question correctly, utilizing a violin plot overlaid with a strip plot.

Choice of Visualization

A Violin Plot overlaid with a Strip Plot is utilized because it provides an effective view of the data distribution, avoiding the loss of spatial alignment that occur when using separate histograms. The visualization is made more effective by plotting both cohorts side by side on a common vertical scale. Because they share the exact same Y-axis, accurate relative judgments regarding time differences can be instantly made. The density of the student population is accurately expressed through the smoothed curve of the violin shape, where a wider area naturally indicates a higher student count at that specific minute mark. Furthermore, clear discriminability is maintained by overlaying the strip plot.

Visual Encoding Breakdown:

- **Marks:** Areas are used as the foundational mark to represent the overall density and statistical distribution of students. Additionally, point marks are used to represent individual student records.
- **Channels:**
  - *Spatial Position (Y-axis):* Used to encode the total time taken in minutes on an aligned, common scale.
  - *Spatial Position (X-axis):* Used to encode the the student cohorts (High Performer and Low Performer).
  - *Size of Violin Width:* Used to encode the student density at a given time.
  - *Color :* Used to encode the categorical identity of the cohorts, leveraging distinct colors (green for high performers, red for low performers).
  - *Color Opacity):* Used on the point marks to visually encode density.

5.5 Hypothesis 5 : There is an optimal time range for achieving higher scores, and students who finish too quickly or take too long tend to score lower

Conclusion: Rejected

The hypothesis is rejected based on the visual and statistical evidence. While an increasing trend can occasionally be observed in first attempt of quiz 2, it is not a consistent or universal pattern. For example, in Quiz 1, the trend is almost flat, indicating that the time spent had no reliable impact on the score. The visualization of quiz 3 also does not show a optimal time range to achieve higher scores. Furthermore, the presence of tall vertical error bars across the time bins indicates considerable variance in the data. Within the same time group, some students scored perfect marks while others scored zero. Therefore, a reliable, universal optimal time range for achieving higher scores does not exist.

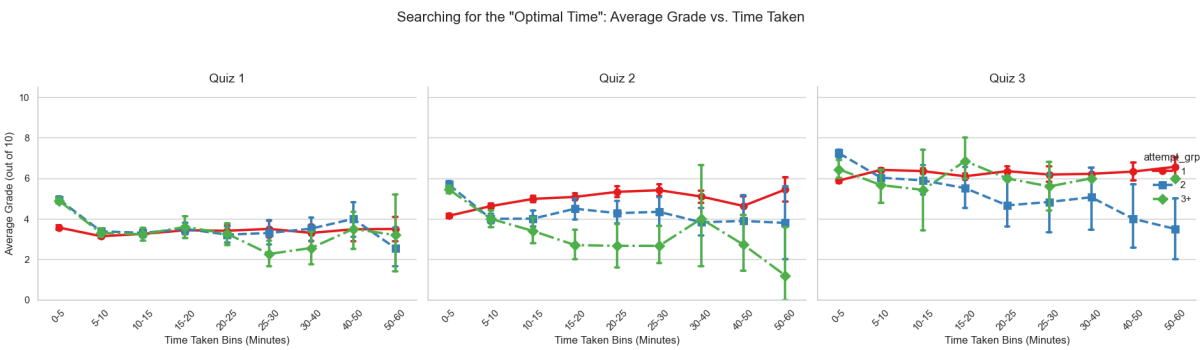

Figure 5: Average grade versus time taken across quizzes, utilizing a faceted mean line chart with 95% confidence interval error bars.

Choice of Visualization

A Faceted Mean Line Chart with Error Bars is used because it effectively manages visual complexity while displaying the statistical rigor of the data. To handle complexity without violating the Discriminability, the data is faceted and juxtaposed into three distinct, side by side panels for the different quizzes. If all lines were plotted on a single graph, severe visual clutter would occur. Accurate relative judgments are supported by forcing all three juxtaposed panels to share the exact same vertical scale on the Y-axis. This allows it to be instantly seen if one quiz is fundamentally easier than another without reading the actual numbers as well. A line is drawn between the averages to instantly perceive the shape of the trend as a single continuous sequence rather than disconnected points. Furthermore, Separability is guaranteed through redundant encoding. Different colors, marker shapes, and line styles are used to represent the different attempts so that even if lines cross, the viewer never loses track of the specific sequence. Finally, the Effectiveness of the visual is enhanced by adding horizontal caps to the vertical error bars, which helps the viewer to measure their length and assess the variance more accurately.

Visual Encoding Breakdown:

- **Marks:** Connection lines are used as the foundational mark to connect data points and illustrate trends over time. Point marks are utilized to explicitly represent the mathematical mean score for each time bin. Additionally, vertical line marks with horizontal caps are used to represent the error bars for 95% confidence intervals.
- **Channels:**
  - *Spatial Position (Y-axis):* Used to encode the average grade on a common, aligned scale.
  - *Spatial Position (X-axis):* Used to encode the sequential time taken, which is categorized into sequential bins.
  - *Color :* Used to encode the categorical identity of the attempt number.
  - *Shape:* Used as a redundant channel to encode the attempt number, ensuring each sequence remains distinct.
  - *Length of Vertical Error Bars:* Used to encode data variance.

Task 2: Additional Hypotheses

After evaluating the five instructor-provided hypotheses, five additional hypotheses were developed from exploratory patterns in retry behaviour, score growth, and cross-quiz performance. The same evidence-based workflow was used to assess these additional claims.

5.6 Task 2 - Hypothesis 1 : Lower first-attempt score is associated with more retries

Conclusion: Accepted

The hypothesis is accepted based on the visualization and descriptive statistics. Students who begin with very low scores retry more often: the median total number of attempts is 2 for first-attempt scores of 0 and 2, but only 1 for first-attempt scores of 4, 6, 8, and 10. The plotted trend declines steadily from left to right, and the Spearman rank correlation between first-attempt score and total attempts is strongly negative ($\rho \approx -0.572$). Therefore, lower initial performance is clearly associated with a greater tendency to retry the quiz.

Figure 6: Relationship between first-attempt score and the median number of total attempts, using a line chart with an interquartile ribbon.

Choice of Visualization

A line chart with point markers and an interquartile ribbon is utilized because the hypothesis concerns how an ordered score scale relates to a summary of retry behaviour. The connected line makes the monotonic trend easy to perceive, while the ribbon preserves information about spread without cluttering the chart with every individual record. This design is especially suitable because the first-attempt score takes a small set of ordered values (0, 2, 4, 6, 8, 10), allowing the viewer to compare the median retry count at each score level directly.

Visual Encoding Breakdown:

- **Marks:** A 1D connection line is used to show the overall trend, point marks show the median number of attempts at each score level, and a shaded area shows the interquartile range.
- **Channels:**
  - *Spatial Position (X-axis):* Used to encode the ordered first-attempt score levels.
  - *Spatial Position (Y-axis):* Used to encode the median total number of attempts.
  - *Area (Ribbon Height):* Used to encode the spread between the 25th and 75th percentiles.
  - *Color (Hue):* A single blue hue is used to maintain focus on the ordered trend rather than introducing unnecessary categorical separation.

5.7 Task 2 - Hypothesis 2 : Attempt 1 to 2 gain is consistently large across quizzes

Conclusion: Rejected

The hypothesis is rejected based on the visualization. A large second-attempt gain is not observed consistently across the three quizzes. Quiz 3 shows a clear median increase from 6 to 8 between Attempts 1 and 2, but Quiz 1 remains flat at a median of 4 across the first several attempts, and Quiz 2 also stays at a median of 4 until Attempt 4. Therefore, while some quizzes benefit from an early retry, the effect is not strong or uniform enough to be considered a consistent pattern across the dataset.

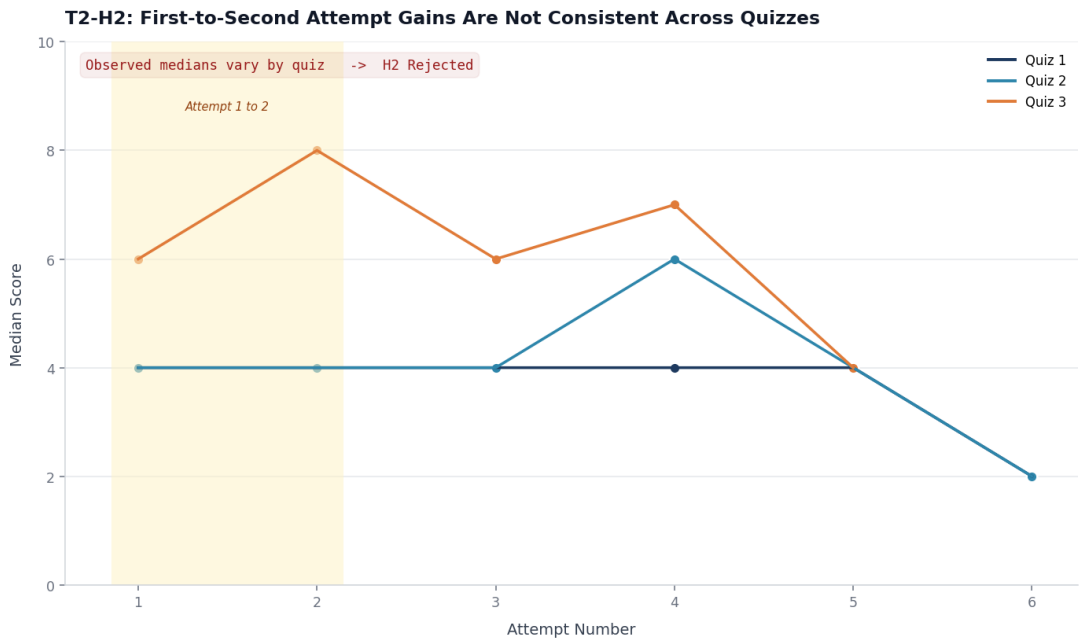

Figure 7: Median score progression by attempt number for each quiz, using a multi-line chart.

Choice of Visualization

A multi-line chart is utilized because the analysis compares the same ordered attempt sequence across multiple quizzes. The line mark highlights change from one attempt to the next, while separate colored trajectories make it easy to determine whether the early improvement pattern replicates across quizzes. Highlighting the Attempt 1 to 2 region further directs attention to the specific part of the curve targeted by the hypothesis.

Visual Encoding Breakdown:

- **Marks:** Connection lines show score trajectories across attempts and point marks identify the exact median score at each attempt.
- **Channels:**

- *Spatial Position (X-axis):* Used to encode attempt number as an ordered sequence.
- *Spatial Position (Y-axis):* Used to encode median score on a common quantitative scale.
- *Color (Hue):* Used to distinguish the three quizzes while preserving comparability across shared axes.
- *Background Shading:* Used as an annotation channel to emphasize the Attempt 1 to 2 interval being evaluated.

5.8 Task 2 - Hypothesis 3 : Longer waiting before retry is linked to smaller improvement

Conclusion: Accepted

The hypothesis is accepted after excluding impossible negative retry gaps caused by inconsistent timestamps in the raw export. The box plot shows that students who retry within one hour achieve the largest improvement, with a median gain of 6 marks, whereas the 1–24 hour, 1–3 day, and more-than-3-day groups all center around a median gain of only 2 marks. After removing invalid negative gaps, the rank correlation between waiting time and score change becomes negative ($\rho \approx -0.221$), indicating that longer delays are generally associated with smaller improvements.

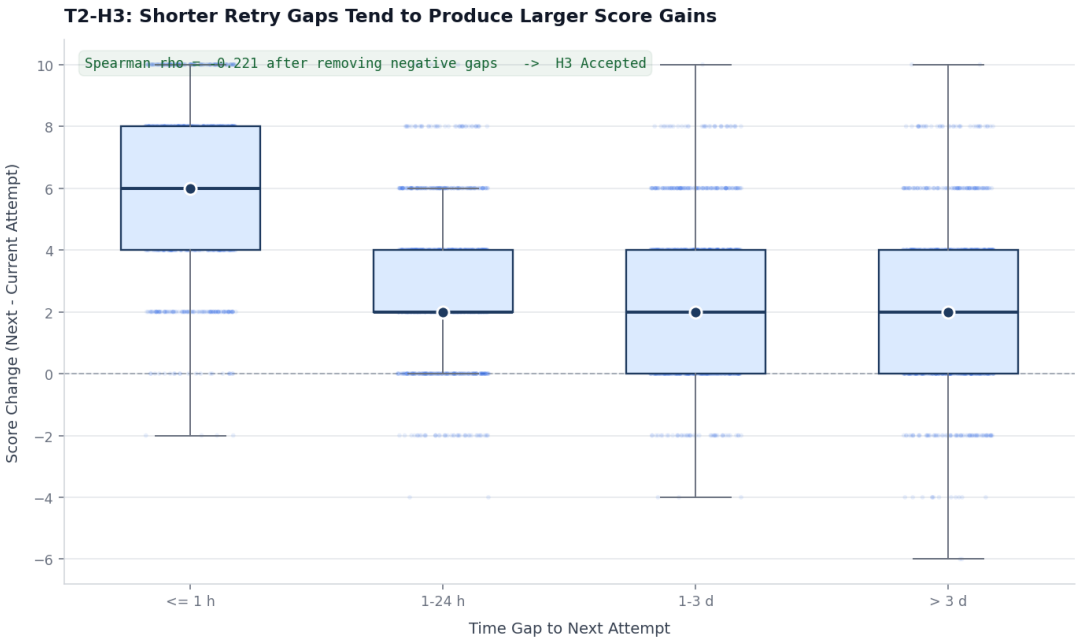

Figure 8: Score improvement versus waiting time before the next attempt, using a box plot with jittered sample points.

Choice of Visualization

A box plot overlaid with jittered points is utilized because the hypothesis concerns the distribution of score changes within a small number of waiting-time buckets. The box plot summarizes medians, quartiles, and spread clearly, while the semi-transparent sample points reveal the underlying density and outliers without overwhelming the viewer. This makes it possible to compare both typical improvement and variability across gap categories.

Visual Encoding Breakdown:

- **Marks:** Box-and-whisker marks summarize each waiting-time group, point marks show sampled individual observations, and larger point marks highlight the group medians.
- **Channels:**
  - *Spatial Position (X-axis):* Used to encode the ordinal waiting-time categories.
  - *Spatial Position (Y-axis):* Used to encode score improvement between the current and next attempt.
  - *Color (Hue):* A consistent blue hue is used so attention remains on distributional differences rather than category identity.
  - *Opacity:* Used on the sampled points to reveal concentration through overlap while preserving outlier visibility.

5.9 Task 2 - Hypothesis 4 : Most 8+ attainment happens in early retries, then plateaus

Conclusion: Accepted

The hypothesis is accepted. The cumulative share of students reaching a score of at least 8 rises from 20.6% on the first attempt to 29.8% by the second attempt and 31.8% by the third. After that, the increase is minimal, reaching only 32.9% by the sixth attempt. This indicates that most high-score attainment happens during the early retries, after which additional attempts provide only marginal benefit.

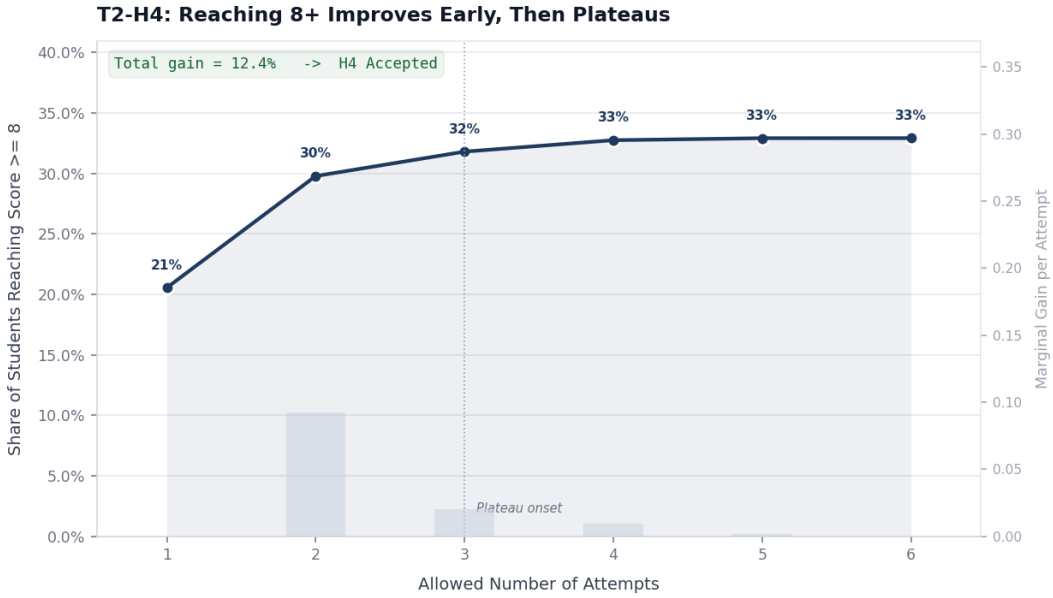

Figure 9: Cumulative share of students reaching 8 or more marks by allowed attempt count, with marginal gains shown as bars.

Choice of Visualization

A cumulative line chart with secondary marginal-gain bars is utilized because the hypothesis has two linked components: overall growth and the rate at which that growth slows down. The line communicates the cumulative attainment trajectory, while the faint bars show how much new attainment is added at each extra attempt. This dual encoding makes the onset of the plateau visually immediate.

Visual Encoding Breakdown:

- **Marks:** A connection line and point marks represent cumulative attainment, while bar marks represent marginal gain per additional attempt.
- **Channels:**
  - *Spatial Position (X-axis):* Used to encode the allowed number of attempts.
  - *Spatial Position (Y-axis):* Used to encode the cumulative share of students reaching at least 8 marks.
  - *Length (Bars):* Used to encode the extra gain contributed by each additional attempt.
  - *Color (Hue):* A dark blue line emphasizes the primary cumulative story, while pale gray-blue bars deliberately recede to the background as supporting evidence.

5.10 Task 2 - Hypothesis 5 : Students perform better on first attempts in later quizzes

Conclusion: Accepted

The hypothesis is accepted based on the cross-quiz comparison of first attempts. The mean first-attempt score increases from 3.37 in Quiz 1 to 4.66 in Quiz 2 and 6.18 in Quiz 3, and the median rises from 4 in the earlier quizzes to 6 in Quiz 3. The upward progression of bar heights, together with the confidence intervals, indicates that first-attempt performance improves across the quiz sequence. This suggests either cumulative learning, increasing familiarity with the quiz format, or both.

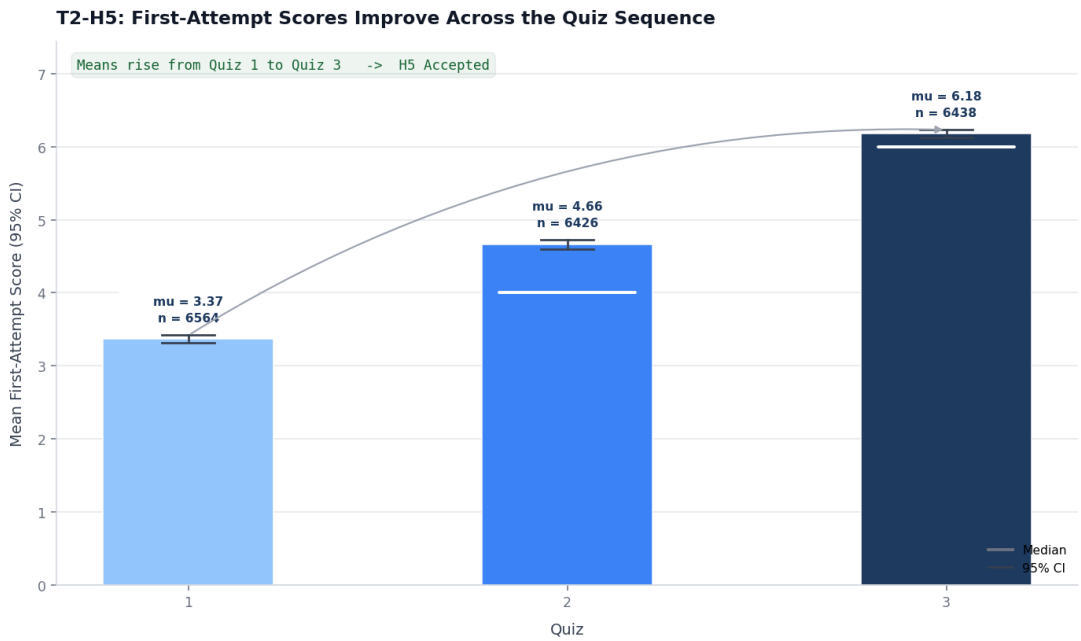

Figure 10: Mean first-attempt score by quiz, using bars with 95% confidence intervals and median markers.

Choice of Visualization

A bar chart with confidence intervals and internal median markers is utilized because the hypothesis requires a clean comparison of first-attempt summary statistics across three discrete quizzes. The bars communicate the mean level clearly, the error bars provide an indication of uncertainty, and the median markers prevent the chart from relying on a single summary measure. The light-to-dark blue progression reinforces the temporal ordering from Quiz 1 to Quiz 3.

Visual Encoding Breakdown:

- **Marks:** Rectangular bars encode mean first-attempt score, vertical error-bar lines encode uncertainty, and short horizontal ticks within the bars encode the median.

- **Channels:**
  - *Spatial Position (X-axis):* Used to encode the quiz sequence.
  - *Spatial Position (Y-axis):* Used to encode mean first-attempt score.
  - *Color (Hue/Value):* A light-to-dark blue progression is used to reinforce the ordered progression from earlier to later quizzes.
  - *Length (Error Bars):* Used to encode the 95% confidence interval around the mean.

6 Discussion

6.1 Summary of Verdicts

Table 3: Hypothesis Verdict Summary

#	Hypothesis (Short)	Key Finding	Verdict
H1	Longer time ⇒ higher score	Moderate time (5–20 min) performs best; very long times correlate with lower scores.	**Rejected**
H2	Some questions are consistently harder	Specific questions (e.g., Q4 in Quiz 1,2 and Q3 in Quiz 3) consistently score lower than others within their respective quizzes across all attempts.	**Accepted**
H3	High performers improve; low performers erratic	High performers experience a sharp downward trend on their second attempt, while low performers demonstrate erratic improvement across subsequent attempts.	**Rejected**
H4	High performers answer hard questions faster	Time distributions for both groups overlap substantially; no consistent speed advantage.	**Rejected**
H5	Optimal time range exists	Time spent does not have a reliable, universal impact on scores. The data shows high variance and often flat trends rather than an inverted-U peak or a consistent correlation.	**Rejected**
T2-H1	Lower first score ⇒ more retries	Students scoring 0–2 initially typically retried twice, while students scoring 4+ usually stopped after one attempt.	**Accepted**
T2-H2	Attempt 1 to 2 gain is consistently large	Only Quiz 3 shows a strong early gain; Quizzes 1 and 2 stay mostly flat across early attempts.	**Rejected**
T2-H3	Longer waiting ⇒ smaller gain	After removing invalid negative gaps, immediate retries show the largest median improvement.	**Accepted**
T2-H4	8+ attainment plateaus after early retries	Most growth toward 8+ happens by Attempts 2–3, after which gains become marginal.	**Accepted**
T2-H5	Later quizzes have better first attempts	First-attempt mean score increases from 3.37 to 4.66 to 6.18 across Quiz 1–3.	**Accepted**

6.2 Interpretation

The original Task 1 analysis shows that raw time spent on a quiz is not, by itself, a reliable route to higher performance. Hypotheses H1 and H5 both reject the idea that simply spending longer produces better results. Instead, students who know the material tend to finish quickly without sacrificing score, while longer durations are more often associated with uncertainty or struggle.

Question difficulty remains the clearest stable structural finding from Task 1. Hypothesis H2 was fully accepted, showing that certain specific items (such as Q4 in Quiz 1 and Q3 in Quiz 3) consistently underperform compared to other questions within the same quiz across repeated attempts. Hypothesis H4 was fully rejected. While per-question speed had to be inferred from total quiz time, the data contradicted the expected behavior - in two out of the three quizzes, high-performing students actually took longer to complete the assessment than lower-performing students when answering the hardest questions correctly.

Task 2 adds a complementary story about retry behaviour. Students with weaker first attempts are more likely to retry, and the benefits of retrying are concentrated early. The share of students reaching at least 8 marks rises sharply by the second and third attempts, then flattens, suggesting that a small number of retries captures most of the attainable improvement.

The retry-gap analysis also suggests that feedback is most effective when acted upon quickly. After excluding impossible negative waiting times from the raw timestamps, immediate retries produce substantially larger gains than delayed retries. At the same time, T2-H2 shows that improvement patterns are not identical across quizzes: some quizzes benefit strongly from an early retry, whereas others do not.

Finally, first-attempt performance improves across the quiz sequence. This may indicate cumulative learning, increasing familiarity with Moodle quiz mechanics, or improved preparation as the course progresses. Together, the ten hypotheses show that item difficulty and retry design matter more than raw completion time alone.

7 Conclusion

This project applied data visualization techniques to a dataset of 31,334 quiz attempt records to test ten hypotheses: the five original hypotheses provided in the assignment and five additional hypotheses formulated from exploratory analysis. Through careful chart selection — including box plots, grouped bar charts, multi-line charts, violin plots, cumulative attainment curves, and confidence-interval bar charts — each claim was evaluated with attention to mark selection, channel effectiveness, and visual interpretability.

Across the original Task 1 hypotheses, question difficulty was the most robust finding. Certain questions consistently scored lower than others, while the timing-based claims were mostly rejected. The hypothesized relationship between hard questions and faster answering speeds for high performers was rejected. Even though the available data recorded total quiz duration rather than per-question response time, the overarching trend showed that high performers often took slightly longer to complete the quizzes than their lower-performing peers.

The additional Task 2 hypotheses revealed clearer patterns in retry behaviour. Lower first-attempt scores lead to more retries, shorter retry gaps tend to yield larger improvements, most attainment of scores ≥ 8 happens within the first three attempts, and first-attempt scores improve across the quiz sequence. Together, these findings suggest that well-designed retry opportunities and prompt feedback can support learning, but the benefit of repeated attempts plateaus quickly.

These results have practical implications for assessment design. Instructors should review persistently difficult questions, avoid assuming that slow completion implies stronger understanding, and consider retry policies that emphasize early reattempts and timely feedback rather than unlimited repetition. Future work could enrich the analysis with per-question timing, clickstream data, and controlled learner-level variables.